

Static stiffness mismatch

The MuJoCo Playground default OP3 stiffness, $k_p = 21.1$, is too soft for the measured physical robot. In a static walk-ready pose held for 5 seconds, the real OP3 kept all joints within roughly 0.1-0.2 degrees, while the default simulator showed substantially larger yielding under gravity. Increasing simulated stiffness towards $k_p = 100$ reduced the discrepancy, though it did not fully match the real robot. Increasing k_p in the actuator block also changes the closed-loop bandwidth (see the companion note *The policy bakes in the physics*), so any stiffness change at this layer has to be paired with a re-export of the policy or, more honestly, a retraining cycle at the new gain.

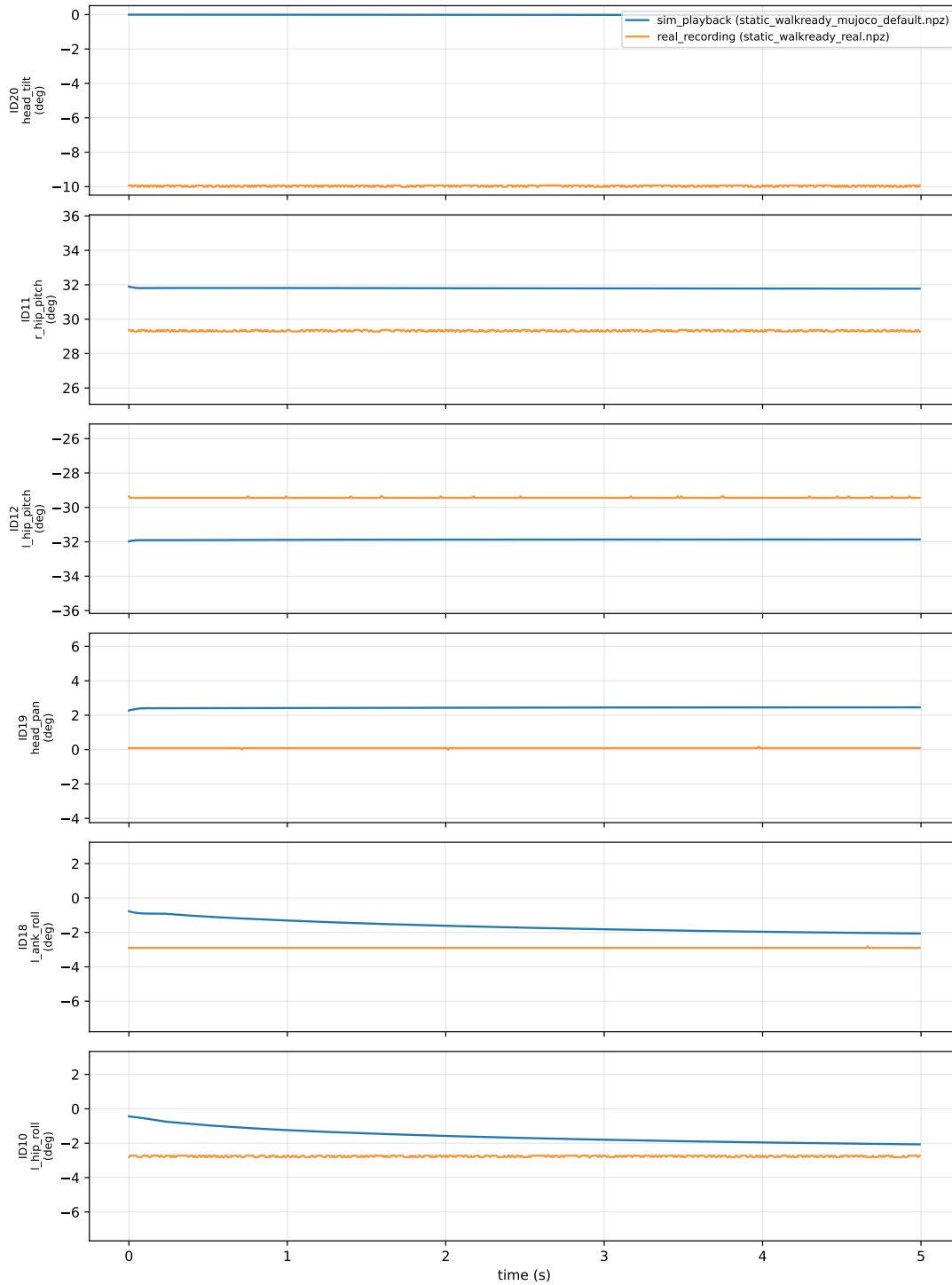


Figure 1: Static walk-ready hold, real OP3 (orange) vs MuJoCo with the Playground default $k_p = 21.1$ (blue). Top six joints by sim-vs-real RMS, in degrees. Every panel covers the same vertical extent (centred on each joint’s own mean) so yield magnitudes are visually comparable. Overall RMS 4.5° across these joints; the real robot holds the pose tightly while the simulator yields visibly under gravity.

The mirror structure between left and right panels is real, and expected. The OP3 has bilateral kinematics, and the walk-ready pose (ROBOTIS page 9) assigns mirror-imaged commanded values to paired joints — e.g. $r_knee \approx -61^\circ$ and $l_knee \approx +61^\circ$, $r_hip_pitch \approx +32^\circ$ and $l_hip_pitch \approx -32^\circ$. Under gravity each leg yields toward “buckle toward straight” in its respective sign convention, so on a per-joint plot the L/R pairs read as vertical mirror images of

each other. This is a useful physical-consistency check: if MuJoCo did *not* show mirror behaviour for symmetric inputs under purely gravitational load, that would point at an asymmetric bug in masses, inertias, or joint axes. The interesting finding here is not the symmetry itself but the absolute yield magnitude — too large with the Playground default k_p .

(Two distinct poses appear in this study and are easy to confuse. The static comparison above uses the ROBOTIS walk-ready pose — the canonical upright hold from the OP3 framework — because that is the pose used for hardware static tests. The training policy on the other hand resets each episode at MuJoCo Playground’s `stand_bent_knees` keyframe, a more deeply-flexed posture that the policy is trained to maintain dynamically. Both are mirror-symmetric between L/R, but the commanded values differ, and the two poses should not be substituted for each other in any rollout comparison.)

Action-page trajectory shape

The ROBOTIS OP3 framework ships with a catalogue of pre-recorded *action pages*: each page is a sequence of joint-position keyframes plus per-step timing, executed on hardware by the action module. Page 204 (`see_rightup`) is a head-turn motion — `head_pan` from $+0.2^\circ$ to $+1.1^\circ$, roughly 41° peak-to-peak — that we use as a deterministic reference signal because it is repeatable and exposes exactly one fast joint (full details of the catalogue and motion file format are in `sde-football-docs/11-reference/robotis-op3-framework/`).

When replaying these action pages in simulation, the softer simulator controller initially appeared to track better than the stiffer one. The cause was an input mismatch rather than a gain mismatch: the simulation was replaying a *linear* interpolation between successive keyframes, while the real ROBOTIS action module generates a *trapezoidal* velocity profile with explicit acceleration and deceleration phases. Porting the trapezoidal generator into the simulation playback path substantially reduced the dynamic tracking error, and the apparent ranking between stiffness settings reversed. The lesson generalises: matching commanded trajectories at the right time-derivative class is at least as important as matching gains.

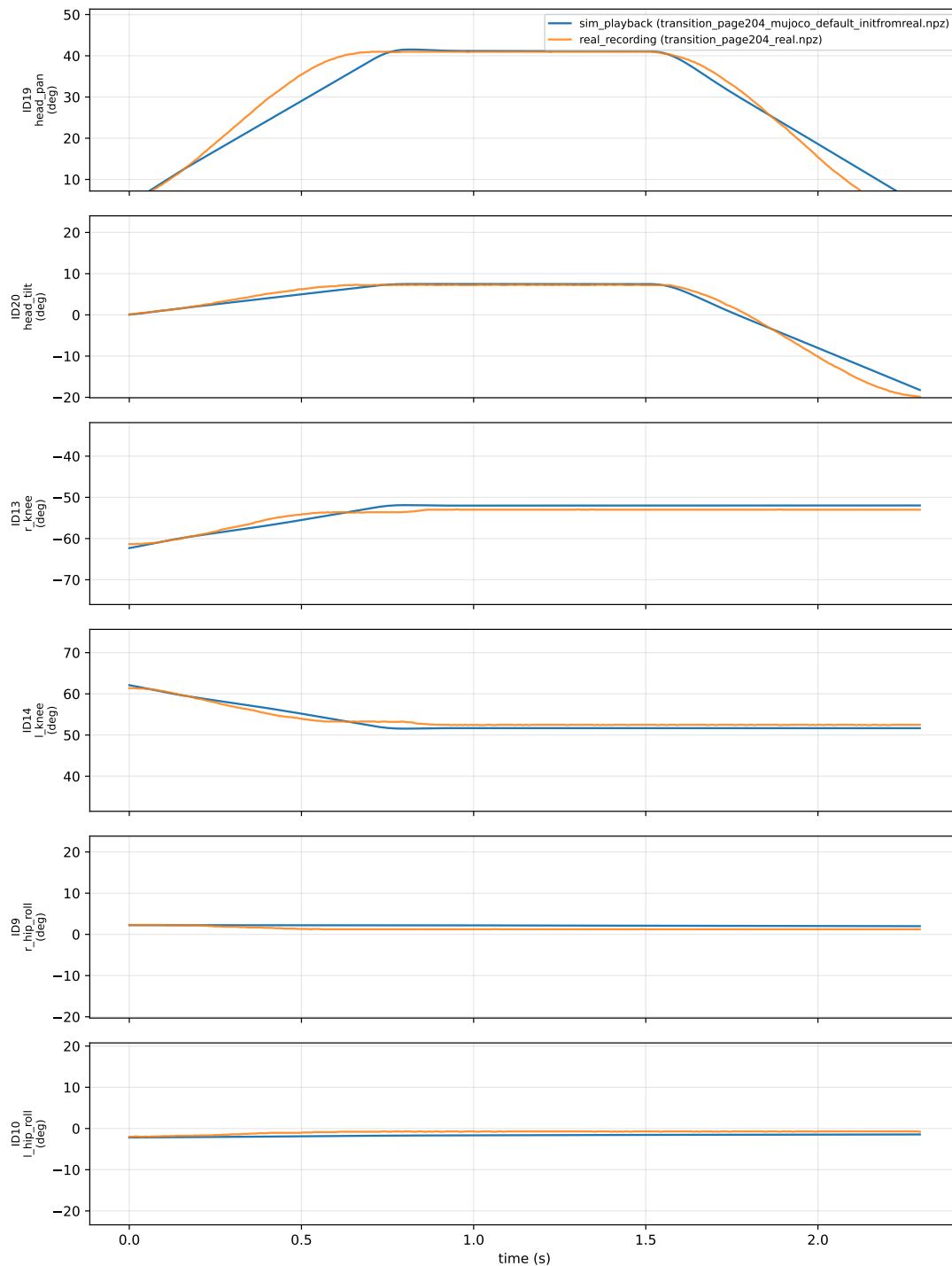


Figure 2: Page 204 (see_rightup) replay. Orange: real OP3 — the trapezoidal profile produced by the on-board action module driving the Dynamixels. Blue: MuJoCo replay with the Playground-default stiffness, initialised from the real robot’s starting pose, with linear interpolation between the same keyframes. Top six joints by sim-vs-real RMS; overall RMS 1.5° across these joints. The residual reflects both the linear-vs-trapezoidal trajectory shape and the static stiffness gap on the shoulder/arm joints.

Foot friction and contact geometry

The physical OP3 has four screws protruding roughly 10~mm below each sole. On natural grass and artificial turf they do not behave like a friction surface at all — they mechanically interlock

with the substrate, and the foot can only lift, not slide. Shuffling gaits that are stable in simulation therefore fail on hardware: the foot catches and the robot stumbles. The training simulator’s flat-plane MuJoCo Coulomb model (default $\mu = 0.8$) does not capture this regime and admits sliding strategies the policy then relies on.

Mapping observed surfaces onto the simulator: $\mu \approx 0.3$ corresponds to wet grass (foot slides easily), $\mu \approx 0.5$ to the Webots RoboCup grass default (moderate grip), $\mu \approx 0.8$ to the MuJoCo training default (good grip but still slides), and $\mu \gtrsim 2.0$ approximates the spiked-foot-on-grass contact where the foot effectively cannot slide. A reasonable domain-randomisation range that brackets the actual hardware deployment surface is $\mu \in [0.5, 3.0]$; a spike-specific configuration would use $\mu \in [2.0, 5.0]$.

For the policy to be robust in this regime it has to learn three distinct things: (i) lift the swing foot high enough to clear the 10~mm screw heads on the stance side; (ii) place the foot cleanly without sideways drag, since lateral motion is not available once the spikes engage; and (iii) turn by stepping rather than pivoting in place, because the spikes prevent in-place foot rotation on the ground. Each maps onto an existing reward term — `feet_clearance`, `feet_slip`, and the body-yaw tracking term respectively — but with weights tuned for sliding-friction surfaces rather than interlock. The full recommendation, including specific weight changes, is in the earlier note `friction-and-domain-randomisation.md`.

Why these matter for the SPECS audit

Each finding maps cleanly onto a layer the audit framework already checks: stiffness at the actuator block, trajectory shape at the deployment-control filter, and foot friction at the contact geometry. Together with the closed-loop bandwidth signature surfaced in this study (companion note *The policy bakes in the physics*), they form a four-item checklist of fidelity gaps to be resolved before any zero-shot deployment claim on OP3 hardware is credible. None of these are new in the literature, but their interaction is specifically what the audit makes legible.